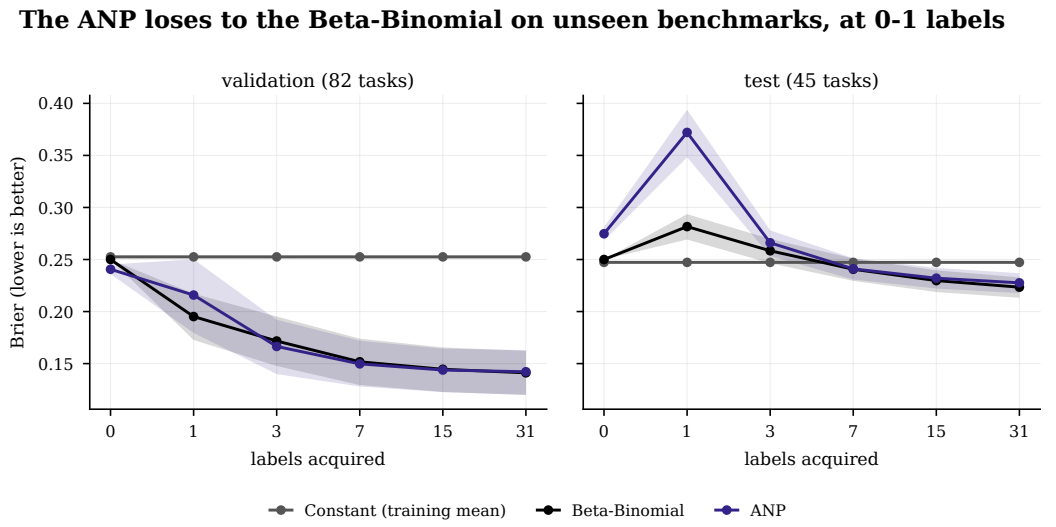



Figure (single-pair). Mean Brier on each task's held-out evaluation items after 0-31 labels, each subject-benchmark task simulated alone on its own labels with random acquisition (validation: 3 candidate orders, test: 10; benchmarks never seen in training). Bands: 95% bootstrap over tasks. The ANP's whole deficit sits at 0 and 1 labels; from 3 labels on it matches the Beta-Binomial. The constant ignores labels and sets the floor. Brier ALC (val / test): Constant (training mean) 0.2525 / 0.2472; Beta-Binomial 0.1717 / 0.2494; ANP 0.1735 / 0.2725. Pooled Rasch needs other pairs' labels and is shown on the multi-pair page.

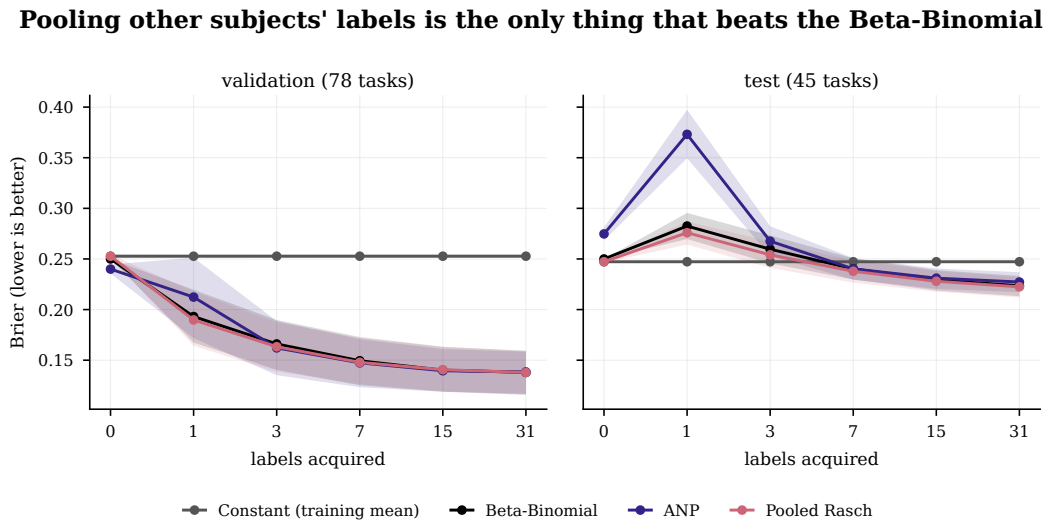


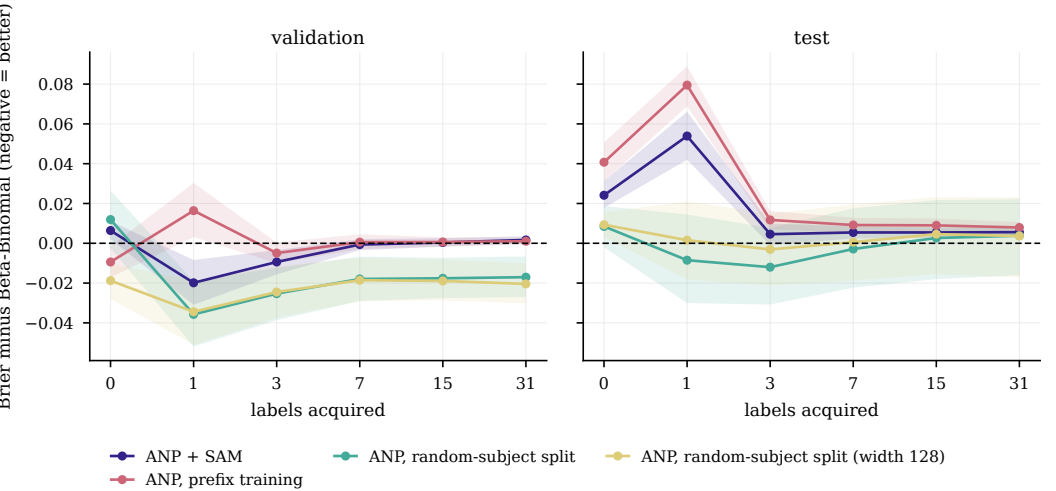

Figure (multi-pair). Same models on CodaBench-like draws: one or two subjects per benchmark, labels acquired in the the online simulation's frozen orders, and every pair's labels available to the predictor (60 validation and 150 test draws). Models that ignore other pairs are placed on the same draws from their single-pair scores. Pooled Rasch fits benchmark rate, subject ability and item difficulty from all pairs' labels and is the only model below the Beta-Binomial on test at every budget. Bands: 95% bootstrap over tasks. Brier ALC (val / test): Constant (training mean) 0.2527 / 0.2472; Beta-Binomial 0.1685 / 0.2497; ANP 0.1701 / 0.2726; Pooled Rasch 0.1674 / 0.2461.

# Only pooling other subjects' labels beats the Beta-Binomial on unseen bench

| model | protocol | val ALC | test ALC | test minus Beta-Binomial [95% CI] |
|---|---|---|---|---|
| **ANP training recipe and split** | | | | |
| ANP + SAM | single-pair | 0.1667 | 0.2663 | +0.0168 [+0.0128, +0.0211]* |
| ANP, prefix training | single-pair | 0.1735 | 0.2762 | +0.0267 [+0.0231, +0.0306]* |
| ANP, random-subject split | single-pair | 0.1519 | 0.2465 | -0.0029 [-0.0209, +0.0154] |
| ANP, random-subject split (width 128) | single-pair | 0.1485 | 0.2515 | +0.0020 [-0.0153, +0.0190] |
| **Combining the ANP with the Beta-Binomia** | | | | |
| ANP | single-pair | 0.1735 | 0.2725 | +0.0230 [+0.0198, +0.0267]* |
| ANP with Beta-Binomial fallback at 0-1 labels | single-pair | 0.1703 | 0.2519 | +0.0025 [+0.0003, +0.0048]* |
| ANP-Beta-Binomial ensemble | single-pair | 0.1702 | 0.2556 | +0.0062 [+0.0050, +0.0074]* |
| ANP-Beta-Binomial ensemble, per-budget weig | single-pair | 0.1688 | 0.2538 | +0.0043 [+0.0027, +0.0062]* |
| ANP + residual correction | single-pair | 0.1936 | 0.2996 | +0.0502 [+0.0465, +0.0537]* |
| Beta-Binomial + residual correction | single-pair | 0.1953 | 0.2986 | +0.0492 [+0.0452, +0.0529]* |
| Raw hit rate | single-pair | 0.1892 | 0.2879 | +0.0385 [+0.0355, +0.0413]* |
| **Hierarchical Beta-Binomial: subject-metad** | | | | |
| Hierarchical Beta-Binomial | single-pair | 0.1703 | 0.2489 | -0.0006 [-0.0027, +0.0017] |
| Hierarchical Beta-Binomial + provider | single-pair | 0.1694 | 0.2452 | -0.0043 [-0.0078, +0.0001] |
| Hierarchical Beta-Binomial + model name, pro | single-pair | 0.1710 | 0.2445 | -0.0049 [-0.0090, +0.0001] |
| Hierarchical Beta-Binomial + provider, release | single-pair | 0.1807 | 0.2626 | +0.0131 [+0.0046, +0.0220]* |
| Hierarchical Beta-Binomial + model name, pro | single-pair | 0.1726 | 0.2455 | -0.0039 [-0.0084, +0.0015] |
| **Pooling all pairs' labels (multi-pair)** | | | | |
| Pooled Rasch | multi-pair | 0.1674 | 0.2461 | -0.0035 [-0.0045, -0.0026]* |
| Pooled Rasch without item difficulty | multi-pair | 0.1675 | 0.2477 | -0.0020 [-0.0029, -0.0011]* |
| Pooled Rasch + provider prior | multi-pair | 0.1647 | 0.2500 | +0.0003 [-0.0027, +0.0033] |
| **Not simulated online** | | | | |
| ANP, budget-ladder training | - | - | - | trained, never simulated online |
| ANP + prior anchor | - | - | - | trained, never simulated online |
| ANP + SAM, budget-ladder training | - | - | - | trained, never simulated online |
| ANP without item embeddings | - | - | - | training; step-400 checkpoint only |

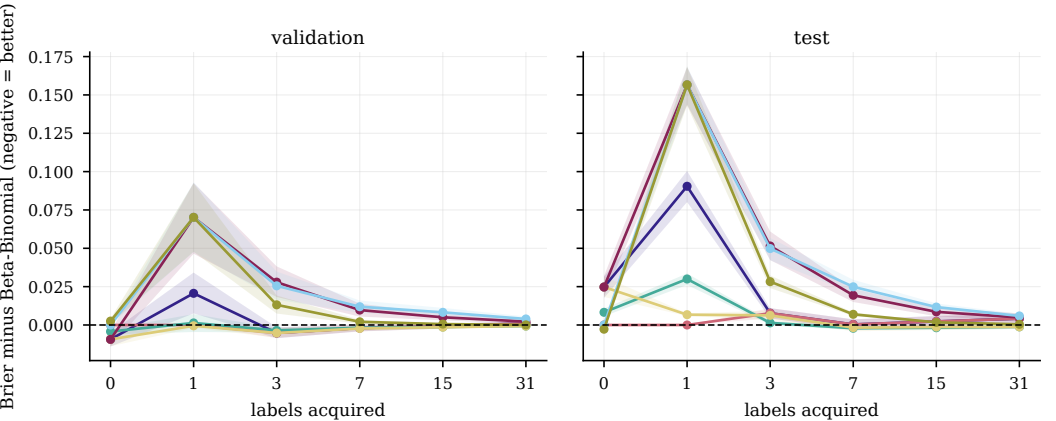
Table. Macro Brier ALC (lower is better) on validation and on the untouched test benchmarks (algotune, cmmlu, editbench), random acquisition. The last column is the paired test difference to the Beta-Binomial (Jeffreys prior) under the same protocol, with a 95% bootstrap interval over tasks; * marks intervals that exclude 0. Single-pair Beta-Binomial: 0.1717 / 0.2494 (val / test); multi-pair: 0.1685 / 0.2497. Validation selected the ensemble weights, the metadata L2 and the pooled-Rasch priors. The random-split ANPs trained on editbench. Rows under 'Not simulated online' were trained but have no online-simulation result.

## No training recipe fixes the ANP; having seen the benchmark does



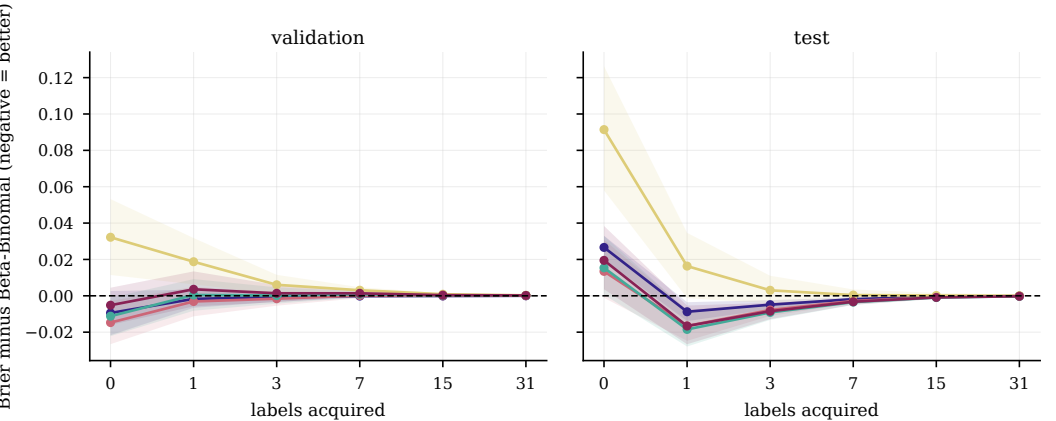

Ablation (single-pair). Paired Brier difference to the Beta-Binomial (Jeffreys prior) at each label budget, validation (left) and test (right); bands: 95% bootstrap over tasks; the dashed line is the Beta-Binomial. SAM and prefix training of the benchmark-holdout ANP stay above the Beta-Binomial on test, by a wide margin at 0-1 labels and slightly from 3 labels on. The random-split ANPs trained on editbench, one of the three test benchmarks; their small test difference averages a large win on editbench with a loss on algotune and cmmlu.

**The Beta-Binomial can limit the ANP's damage, but not turn it into a gain**



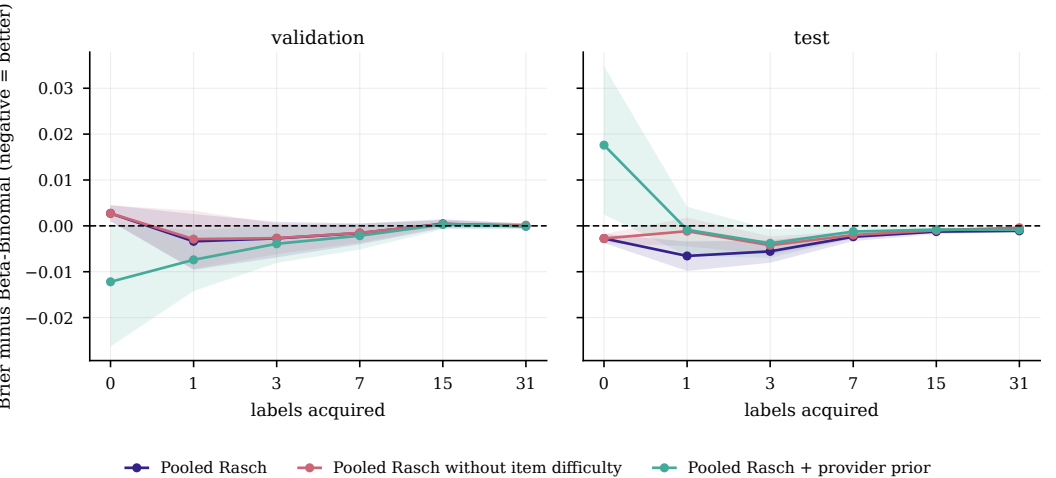

Ablation (single-pair). Paired Brier difference to the Beta-Binomial (Jeffreys prior) at each label budget, validation (left) and test (right); bands: 95% bootstrap over tasks; the dashed line is the Beta-Binomial. The ANP combined with the Beta-Binomial: the fallback uses the Beta-Binomial below two labels, the ensemble averages the two predictions with weights fitted on validation and frozen for test (the validation curves for the ensembles are in-sample). The residual-correction (AIPW-style) variants shift each prediction by the mean residual of the labels seen so far; after one label that shift is about +/-0.5, which is why they spike at budget 1. The raw hit rate shows the same spike for the same reason: no shrinkage.

## Subject metadata barely helps, and release date hurts




Ablation (single-pair). Paired Brier difference to the Beta-Binomial (Jeffreys prior) at each label budget, validation (left) and test (right); bands: 95% bootstrap over tasks; the dashed line is the Beta-Binomial. A Beta-Binomial with a prior fitted on training benchmarks, whose mean depends on the subject's provider, model name and release date as sent by CodaBench. Test models are newer than any training model, so the release-date term extrapolates and hurts at budget 0. A leave-one-benchmark-out check on the 14 non-test benchmarks did not confirm the small gains of the other arms.

## Other subjects' labels help; a provider prior undoes it




Ablation (multi-pair). Paired Brier difference to the Beta-Binomial (Jeffreys prior) at each label budget, validation (left) and test (right); bands: 95% bootstrap over tasks; the dashed line is the Beta-Binomial. Multi-pair draws (one or two subjects per benchmark). Pooled Rasch fits benchmark rate, subject ability and item difficulty from every pair's labels; removing the item term halves the gain. Centring ability on the subject's provider was chosen on validation but loses on test, at budget 0, because test models are stronger than training models from the same providers.

# Test: Brier ALC by model and acquisition policy

algotune, cmmlu, editbench: no model trained on or was selected with these. 45 tasks x 10 order seeds.

Macro-task Brier ALC, lower is better. Bold: lowest among rows that never trained on these benchmarks.

| model | Random | Uncertainty | Confidence extrem | Bernoulli varia | Expected Brier reducti |
|---|---|---|---|---|---|
| Constant (training mean) | **0.2472** | **0.2472** | **0.2472** | **0.2472** | **0.2472** |
| Raw hit rate | 0.2879 | 0.2881 | 0.2855 | 0.2887 | 0.2860 |
| Beta-Binomial | 0.2494 | 0.2494 | 0.2478 | 0.2496 | 0.2474 |
| ANP | 0.2725 | 0.2710 | 0.2754 | 0.2775 | 0.2727 |
| ANP, prefix training | 0.2762 | 0.2732 | 0.2835 | 0.2787 | 0.2782 |
| ANP + SAM | 0.2663 | 0.2642 | 0.2718 | 0.2601 | 0.2651 |
| ANP, random-subject split* | 0.2465 | 0.2445 | 0.2471 | 0.2438 | 0.2522 |
| ANP, random-subject split (width | 0.2515 | 0.2505 | 0.2505 | 0.2544 | 0.2575 |

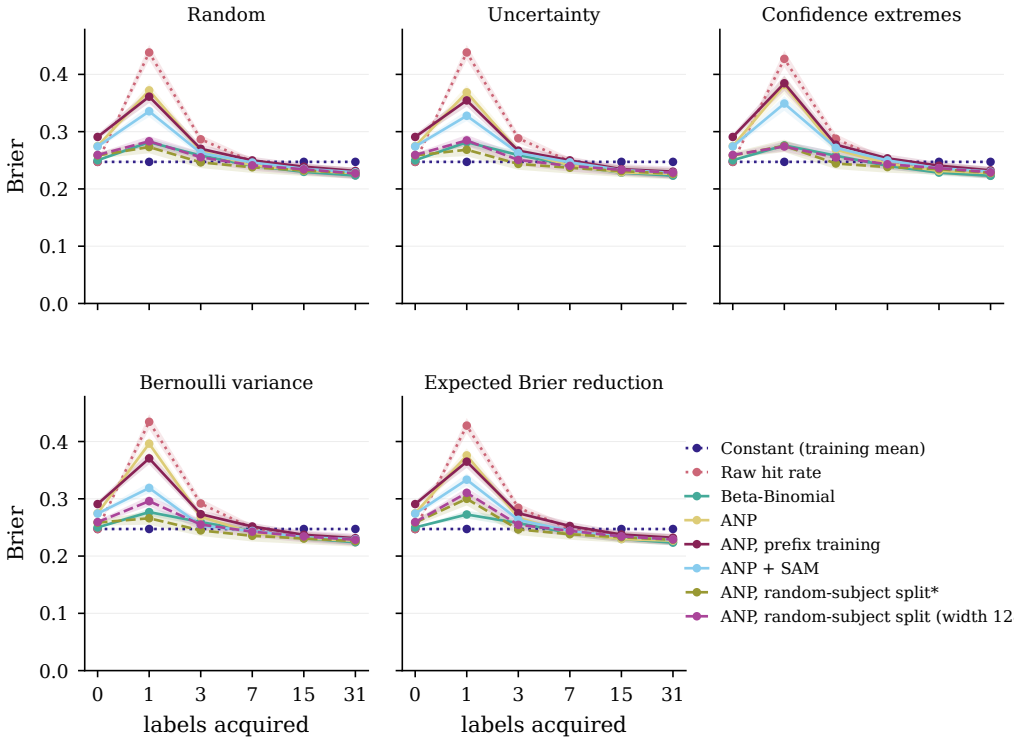
ALC minus Beta-Binomial under the same policy, paired over tasks: mean ± 95% CI. Bold: CI excludes 0.

| model | Random | Uncertainty | Confidence extremes | Bernoulli variance | Expected Brier reductio |
|---|---|---|---|---|---|
| Constant (training mean) | -0.0022 ±0.0088 | -0.0022 ±0.0091 | -0.0006 ±0.0098 | -0.0024 ±0.0091 | -0.0001 ±0.0099 |
| Raw hit rate | **+0.0385 ±0.0030** | **+0.0387 ±0.0030** | **+0.0376 ±0.0033** | **+0.0390 ±0.0053** | **+0.0386 ±0.0040** |
| ANP | **+0.0230 ±0.0034** | **+0.0216 ±0.0033** | **+0.0276 ±0.0086** | **+0.0279 ±0.0059** | **+0.0253 ±0.0056** |
| ANP, prefix training | **+0.0267 ±0.0039** | **+0.0238 ±0.0039** | **+0.0356 ±0.0078** | **+0.0291 ±0.0062** | **+0.0308 ±0.0062** |
| ANP + SAM | **+0.0168 ±0.0042** | **+0.0148 ±0.0045** | **+0.0240 ±0.0075** | **+0.0105 ±0.0059** | **+0.0177 ±0.0066** |
| ANP, random-subject split* | -0.0029 ±0.0187 | -0.0049 ±0.0197 | -0.0007 ±0.0196 | -0.0058 ±0.0195 | +0.0048 ±0.0192 |
| ANP, random-subject split (width | +0.0020 ±0.0177 | +0.0011 ±0.0191 | +0.0027 ±0.0188 | +0.0047 ±0.0191 | +0.0101 ±0.0176 |

* random-subject split: trained on ai2d_test, bird_sql (validation) and editbench (test). Other ANPs used a benchmark holdout.

## Test: Brier by labels acquired (mean ± 1 SE)

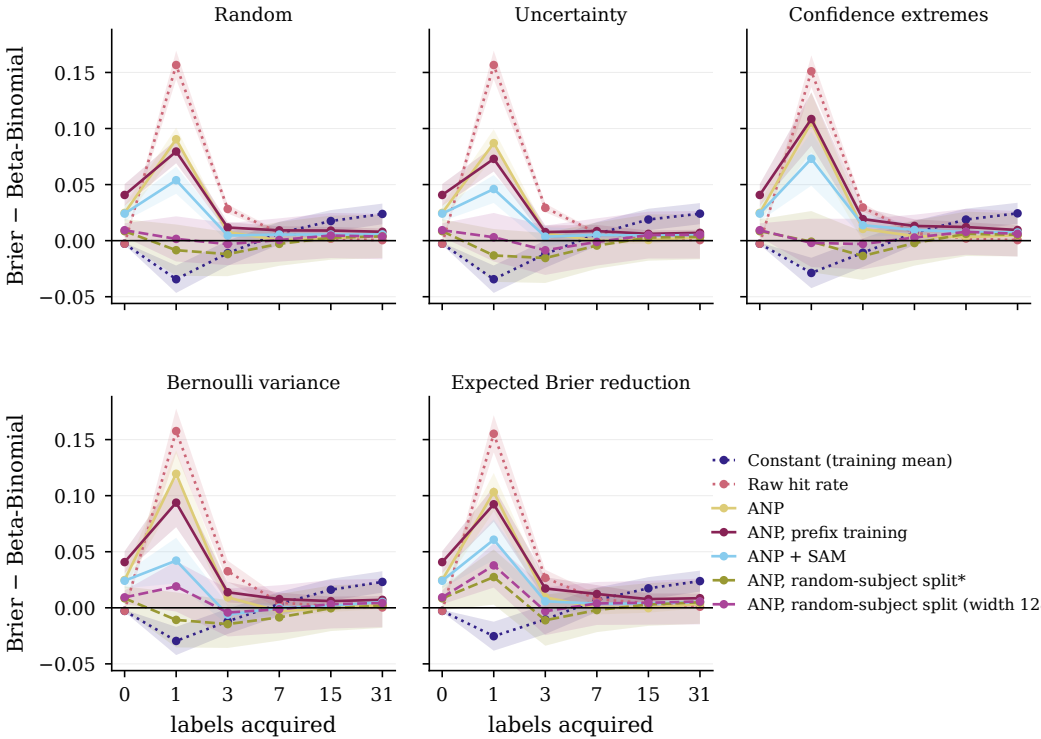
algotune, cmmlu, editbench: no model trained on or was selected with these. 45 tasks x 10 order seeds.




* random-subject split: trained on ai2d_test, bird_sql (validation) and editbench (test). Other ANPs used a benchmark holdout.

## Test: Brier minus Beta-Binomial (paired, 95% CI)

algotune, cmmlu, editbench: no model trained on or was selected with these. 45 tasks x 10 order seeds.




* random-subject split: trained on ai2d_test, bird_sql (validation) and editbench (test). Other ANPs used a benchmark holdout.

## Validation: Brier ALC by model and acquisition policy

ai2d_test, bird_sql, financebench. ANP checkpoints were selected here, which flatters them. 82 tasks x 3 seeds.

Macro-task Brier ALC, lower is better. Bold: lowest among rows that never trained on these benchmarks.

| model | Random | Uncertainty | Confidence extrem | Bernoulli varia | Expected Brier reducti |
|---|---|---|---|---|---|
| Constant (training mean) | 0.2525 | 0.2525 | 0.2525 | 0.2525 | 0.2525 |
| Raw hit rate | 0.1892 | 0.1906 | 0.1944 | 0.1883 | 0.1921 |
| Beta-Binomial | 0.1717 | 0.1726 | 0.1747 | 0.1704 | 0.1731 |
| ANP | 0.1735 | 0.1740 | 0.1759 | 0.1768 | 0.1749 |
| ANP, prefix training | 0.1735 | 0.1748 | 0.1718 | 0.1729 | 0.1719 |
| ANP + SAM | **0.1667** | **0.1679** | **0.1677** | **0.1684** | **0.1708** |
| ANP, random-subject split* | 0.1519 | 0.1538 | 0.1536 | 0.1540 | 0.1611 |
| ANP, random-subject split (width | 0.1485 | 0.1503 | 0.1521 | 0.1528 | 0.1511 |

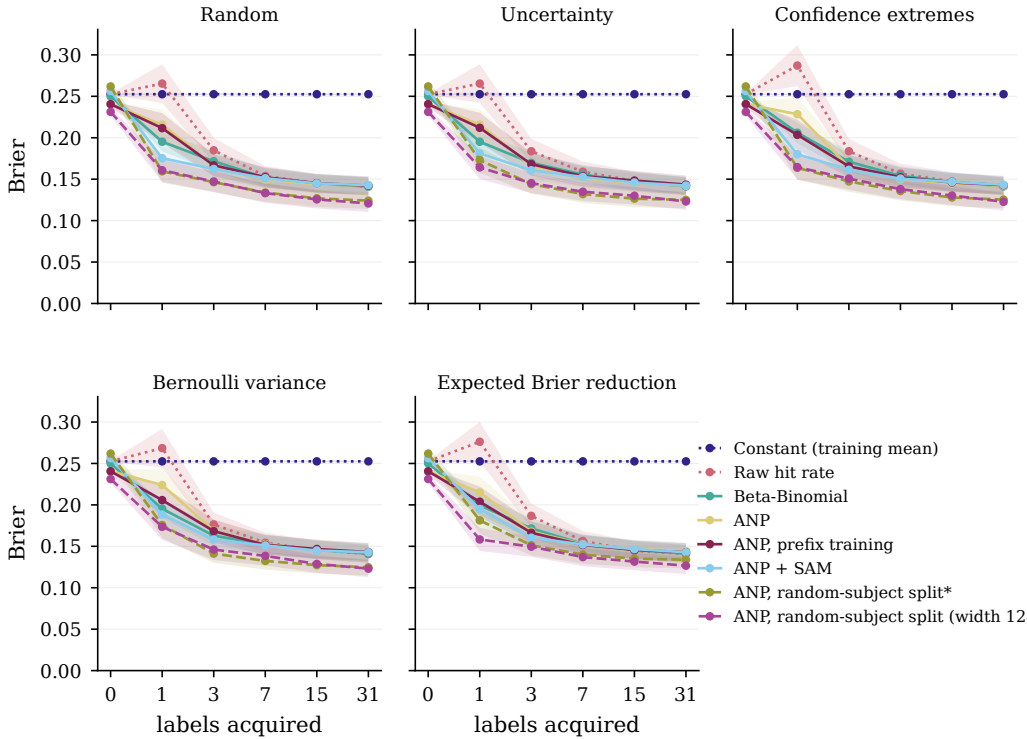
ALC minus Beta-Binomial under the same policy, paired over tasks: mean ± 95% CI. Bold: CI excludes 0.

| model | Random | Uncertainty | Confidence extremes | Bernoulli variance | Expected Brier reductio |
|---|---|---|---|---|---|
| Constant (training mean) | **+0.0808 ±0.0211** | **+0.0799 ±0.0213** | **+0.0779 ±0.0214** | **+0.0822 ±0.0210** | **+0.0795 ±0.0212** |
| Raw hit rate | **+0.0175 ±0.0056** | **+0.0180 ±0.0056** | **+0.0197 ±0.0058** | **+0.0180 ±0.0069** | **+0.0191 ±0.0088** |
| ANP | +0.0017 ±0.0034 | +0.0014 ±0.0034 | +0.0012 ±0.0043 | **+0.0064 ±0.0060** | +0.0018 ±0.0064 |
| ANP, prefix training | +0.0017 ±0.0044 | +0.0022 ±0.0049 | -0.0029 ±0.0067 | +0.0026 ±0.0069 | -0.0012 ±0.0066 |
| ANP + SAM | **-0.0051 ±0.0039** | **-0.0047 ±0.0043** | **-0.0070 ±0.0050** | -0.0019 ±0.0057 | -0.0023 ±0.0054 |
| ANP, random-subject split* | **-0.0198 ±0.0113** | **-0.0188 ±0.0115** | **-0.0211 ±0.0119** | **-0.0164 ±0.0119** | **-0.0120 ±0.0104** |
| ANP, random-subject split (width | **-0.0232 ±0.0110** | **-0.0223 ±0.0107** | **-0.0226 ±0.0115** | **-0.0176 ±0.0128** | **-0.0220 ±0.0093** |

* random-subject split: trained on ai2d_test, bird_sql (validation) and editbench (test). Other ANPs used a benchmark holdout.

## Validation: Brier by labels acquired (mean ± 1 SE)

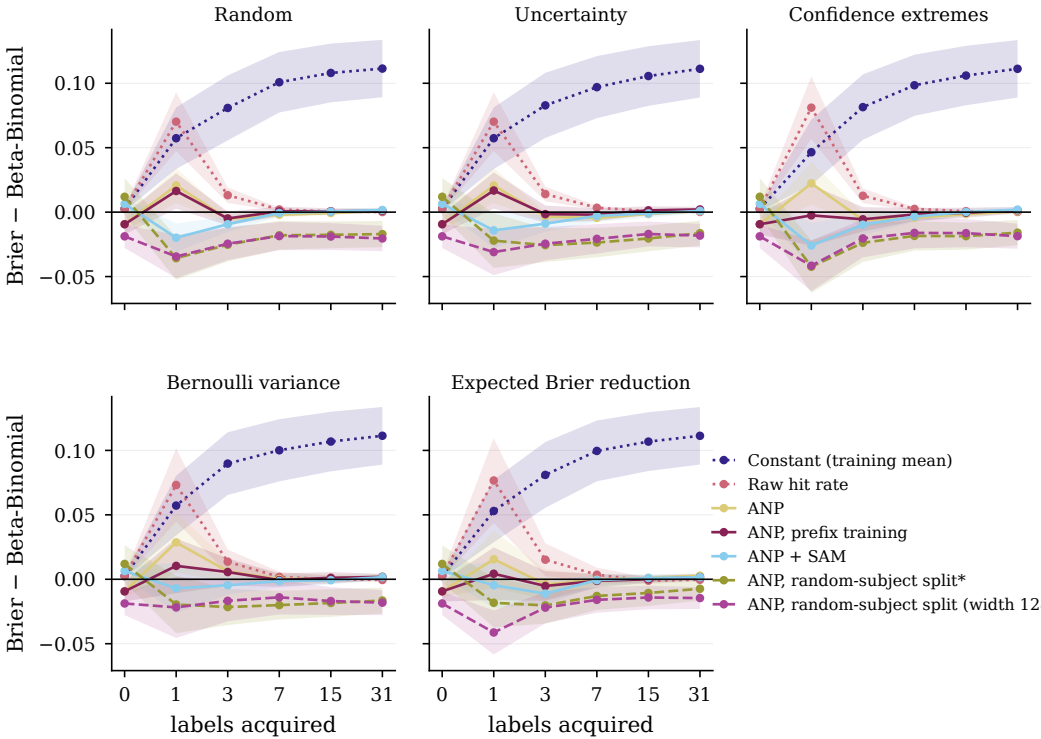
ai2d_test, bird_sql, financebench. ANP checkpoints were selected here, which flatters them. 82 tasks x 3 seeds.




* random-subject split: trained on ai2d_test, bird_sql (validation) and editbench (test). Other ANPs used a benchmark holdout.

## Validation: Brier minus Beta-Binomial (paired, 95% CI)

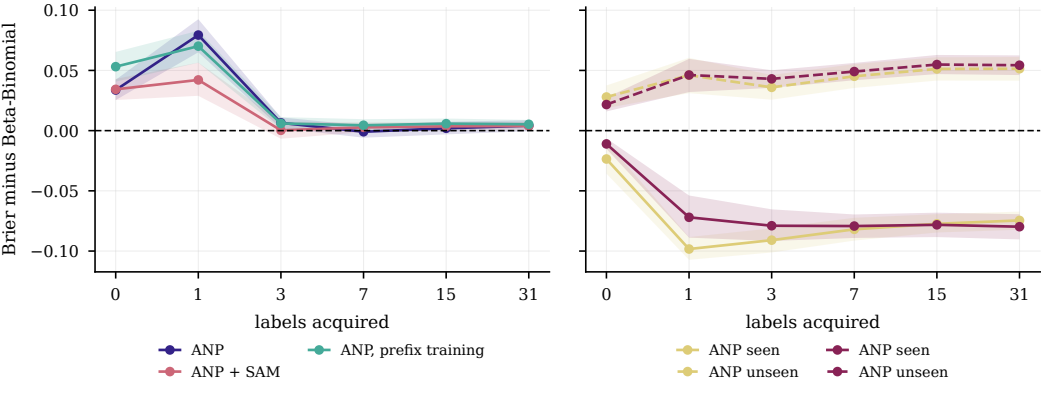
ai2d_test, bird_sql, financebench. ANP checkpoints were selected here, which flatters them. 82 tasks x 3 seeds.




* random-subject split: trained on ai2d_test, bird_sql (validation) and editbench (test). Other ANPs used a benchmark holdout.

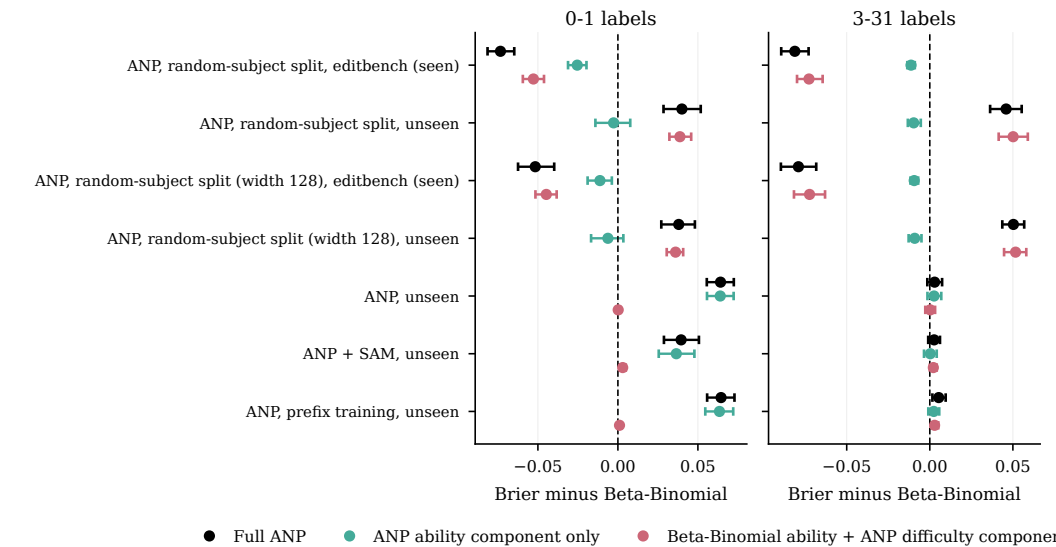
## Where the ANP loses: Brier minus Beta-Binomial at each budget (test)

Left: benchmark-holdout ANPs on the three unseen test benchmarks. Right: random-split ANPs on editbench, which they trained on (solid), and on algotune + cmmlu, which they did not (dashed).



## Which part of the ANP fails on a new benchmark: ability or difficulty? (test)

Ability component: the mean of the ANP's predictions for the task. Difficulty component: each item's offset from that mea
Brier minus Beta-Binomial; negative = better. Bars: 95% task bootstrap.



## Does the ANP's loss grow with subject ability?

One point per task. Ability: the subject's accuracy on the task's acquisition pool (never scored).
Shaded: 5-95% of training-task accuracy (0.00-0.87). No test or validation subject appears in training.

